科研课堂课程作业

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题目 1. 验证 KdV 方程 $u_t + 6uu_x + u_{xxx} = 0$ 的 Lax 对.

解答. 考虑如下两个线性算子 L, A

$$\begin{cases} L = \frac{\partial^2}{\partial x^2} + u \\ A = -4\frac{\partial^3}{\partial x^3} - 6u\frac{\partial}{\partial x} - 3u_x \end{cases}$$

那么我们有

$$\begin{cases} L\varphi = \lambda\varphi \\ \varphi_t = A\varphi \end{cases}$$

不难发现

$$\begin{aligned} &L_t\varphi + L\varphi_t = \lambda\varphi_t = \lambda A\varphi \\ &\Rightarrow L_t\varphi = (AL - LA)\varphi \end{aligned}$$

即 Lax 对满足

$$L_t = AL - LA \equiv [A, L]$$

我们有

$$\begin{aligned} &L_t = u_t \\ &AL = -(4\frac{\partial^5}{\partial x^5} + 10u\frac{\partial^3}{\partial x^3} + 15u_x\frac{\partial^2}{\partial x^2} + 4u_{xxx} + 12u_{xx}\frac{\partial}{\partial x} + 6u^2\frac{\partial}{\partial x} + 9uu_x) \\ &LA = -(4\frac{\partial^5}{\partial x^5} + 12u_{xx}\frac{\partial}{\partial x} + 15u_x\frac{\partial^2}{\partial x^2} + 10u\frac{\partial^3}{\partial x^3} + 6u^2\frac{\partial}{\partial x} + 3u_{xxx} + 3uu_x) \\ &AL - LA = -u_{xxx} - 6uu_x \end{aligned}$$

从而有

$$u_t + 6uu_x + u_{xxx} = 0$$

此即所给 KdV 方程.

题目 2. 验证 KdV 方程 $u_t + 6uu_x + u_{xxx} = 0$ 的 DT.

解答. 略.